

Wilson Gichuhi

Systems & Reliability Engineer

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Professional Summary

Reliability Engineer for AI-Driven Workflows. I architect the deterministic control layer: state machines, event logs, and evaluation invariants that turn probabilistic AI into auditable, recoverable business workflows. I exist because “mostly works” in production is still failure when money, compliance, or trust is on the line.

Professional Experience

AI Evaluation Engineer (Contractor), *Bespoke Labs* [🔗](#)

03/2026

- Contributed task authorship to OpenThoughts, Bespoke Labs' open reasoning dataset initiative that produced state-of-the-art small reasoning models trained on curated chain-of-thought data in collaboration with Stanford, Berkeley, and UT Austin.
- Designed and authored Terminal Bench style agentic coding tasks in the Harbor format, covering concurrency bugs, semaphore leaks, and write-ordering failures across real-world TypeScript and healthcare system scenarios, with full pipeline ownership from environment setup to oracle verification.
- Applied Bespoke-MiniCheck for claim-level factual grounding in evaluation pipelines, using NLI-based verification over LLM-as-judge alone to improve signal reliability in automated test suites.

Builder & AI Systems Engineer, *Promco*(Founder)

03/2026

- Building Promco, an order-to-approval system for liquor distributors that turns WhatsApp orders into commercially reviewable drafts before stock, margin, promo, and compliance decisions are committed.
- Designed the orchestration layer with explicit state machines and durable session handling, replacing free-form agent loops with traceable, recoverable workflow execution for compliance-sensitive order flows.
- Designed the AI guard layer that prevents a WhatsApp order from triggering an irreversible tax submission or a margin-destroying promo without explicit human approval. Enforced these invariants through typed tool interfaces, versioned prompts, and full failure tracing, making the system auditable and resilient against duplicate events, timeouts, and ambiguous inputs common in African mobile commerce.
- Developed the product-resolution layer for noisy real-world order language(Kenyan slang), combining retrieval, ranking, and structured validation to improve SKU matching on colloquial WhatsApp orders.
- Built MCP-based integrations and runtime services over downstream commerce/compliance systems so models operate against live business context rather than ungrounded text generation.

AI Training & Model Evaluation Engineer, AfterQuery

08/2025

- Frontier Model Benchmarking: Conducted systematic evaluations of frontier LLMs for logic correctness and security; engineered automated test suites to identify vulnerabilities across domains.
- SWE-Bench Style Engineering (Project Anvil): Built end-to-end evaluation environments using base snapshots and gold-patch verification; automated oracle verification pipelines for complex domains including HL7 health-tech integrations and secure build systems.
- Terminal Bench v2: Built real-world agentic coding tasks targeting concurrency bugs, security vulnerabilities, and infra debugging — with full pipeline ownership from gold patches to oracle verification and pytest suites in containerized environments.
- Adversarial Engineering: Designed stress-tests for LLM-driven API design, moving beyond naive prompting to automated red-teaming of model-generated system architectures.

Backend Engineer, Alcora (FMCG Operations Platform)

02/2025 – 09/2025

- Architected Headless ERP as a multi-tenant backend with real-time stock and demand intelligence across distributed FMCG actors.
- Built core payment rails with automated debt reconciliation across brands, distributors, and retailers.
- Embedded geospatial intelligence for delivery tracking, route batching, and dispatch automation.
- Enabled promo cost vs incremental gross profit tracking (North Star Metric ≤ 0.6), reducing promotional wastage by 40%.

Founding Engineer, AfyaTelemed

05/2024 – 07/2025

- Architected a HIPAA-compliant telemedicine platform from scratch, covering triage, consultations, pharmacy, and lab integration.
- Built secure video consultation with E2E encryption (WebRTC) and payment processing across local and international gateways.
- Stack: Python/FastAPI, Next.js, WebRTC, AWS

Software Engineer, McSystems & Medical Inc.

04/2024 – 10/2024

- Architected a CRM with voice-to-CRM logging, reducing manual data entry by 70%.
- Rebuilt a legacy PHP medical sales marketplace into an offline-first Next.js platform.
- Designed a unified API layer integrating CRM and marketplace systems.

Fullstack Developer, JHUB Innovation Africa

10/2023 – 05/2024

- Built digital trade and climate-smart agriculture platforms within a Microsoft-partnered innovation ecosystem
- Delivered enterprise-grade authentication and workflow integrations across multi-stakeholder systems.

Technical Skills

Cognitive Architecture & Agentic Systems

- **State-Driven Logic:** XState (Finite State Machines), Apache Burr, Model Context Protocol (MCP).
- **Orchestration Patterns:** Agentic RAG, Multi-Agent Coordination, Tool-use, 12-Factor Agents.
- **Reliability Layer:** Effect-TS (Errors-as-Data), Circuit Breakers, Event-Driven Architectures.

Retrieval Diagnostics & Information Retrieval (IR)

- **Hybrid Search:** Vector + BM25, Rerankers, Atomic Fact Extraction
- **Data Stores:** Qdrant, Pinecone, pgvector (PostgreSQL), DuckDB.
- **Search Optimization:** Causal reasoning for retrieval drift, query expansion, and semantic caching.

UX & Decision Interfaces

- Review dashboards
- Offline-first PWAs (React, TanStack)
- HITL approval surfaces
- Generative UI, Shared Autonomy Interfaces, and Headless UI primitives

Education

Bsc. Computer Technology, JKUAT University

Evals & System Governance

- **Diagnostic Frameworks:** DeepEval, RAGAS, NLI (Natural Language Inference).
- **Safety & Alignment:** LLM-as-a-Judge, Automated Hallucination Detection, Evals-as-Code.
- **Synthetic Data & Robustness:** Curator (Synthetic Data Generation) for edge-case discovery, adversarial stress testing, and dataset augmentation.

Deployments, Model Serving & Resilience

- **Vitals & Optimization:** PagedAttention, Quantization (GGUF, AWQ, EXL2).
- **Resilient Compute:** Docker, Kubernetes (K8s), IaC, Cloudflare Workers (Edge Inference), Ollama.

Browser Automation & Web Agents

- Chrome DevTools Protocol (CDP), Playwright, Stagehand
- Agent action recording, session replay, DOM change detection
- Fault-tolerant navigation in auth flows and dynamic interfaces

2018 – 2023